Combustion, Explosion, and Shock Waves, Vol. 43, No. 2, pp. 212–218, 2007

Heat of Explosion of Commercial and Brisant High Explosives

V. I. Pepekin[1] and S. A. Gubin[1]

UDC 662.217; P = IV or P = IV^(½)

Translated from *Fizika Goreniya i Vzryva*, Vol. 43, No. 2, pp. 100–107, March–April, 2007. Original article submitted April 4, 2005; revision submitted July 5, 2006.

Methods for determining the heat of explosion of high explosives (HEs) with ideal and nonideal processes of explosive decomposition are considered. It is shown that the heat of explosion is of significance for estimating the efficiency of commercial HEs and is used in the energetic characterization of the working capacity. The heat of explosion of brisant HEs is only part of the blast heat of explosion and is the heat content of gaseous detonation products during their isentropic expansion from the initial state to a certain expansion ratio (determined by experimental conditions). The heat of explosion can be obtained by thermodynamic calculations based on physically justified equations of state for fluids (gaseous detonation products in the chemical-reaction zone of the detonation wave in the supercritical state) and condensed nanocarbon phases (nanographite, nanodiamond, and liquid carbon). Experimental and calculated values of the heat of explosion are given. The thermodynamic calculation is inapplicable to commercial HEs because of the nonideal nature of their detonation. The heat of explosion of commercial HEs can be calculated using the Hess law. The heat of explosion of brisant HEs is not a measure of power. The power of HEs is characterized by the propellant performance. It is shown that even detonation velocity cannot be a measure of the power of HEs. The power and detonation parameters of brisant HEs are determined by the energy release density in unit volume of the chemical-reaction zone of the detonation wave and by the rate of energy release from the shock front rather than by the heat of explosion, which cannot be considered a universal characteristic.

Key words: heat of explosion, commercial and brisant HEs, thermodynamics.

EXPERIMENTAL METHODS OF DETERMINING THE HEAT OF EXPLOSION OF COMMERCIAL HE

The extensive use of explosive compositions in the 1930s and 1940s in industry motivated the necessity of developing experimental methods for determining the heats of explosion and the volume of gaseous products of explosive decomposition. The experimentally deter-mined characteristics (heat of explosion and gas vol-ume) of commercial high explosives (HEs) served as grounds for their application. These characteristics are also important today since they are used as the basis for estimating the working capacity (blast effect) of real commercial HEs. By the modern scientific nomen-clature, commercial HEs (usually based on ammonium nitrate) are classified as explosive systems with a non-ideal detonation process. The significance of the heat of explosion and the volume of gaseous products for the estimation of the applicability of commercial HEs has led to the necessity of developing experimental methods for their determination.

For direct determination, facilities of large dimen-sions and increased working capacity have been de-signed and used that provide ignition of large charges with a blasting cap.

To determine the heat of explosion, Apin and Belyaev [1] used a thick-walled bomb with an inner vol-

[1]Semenov Institute of Chemical Physics, Moscow 117977; kors@polymer.chph.ras.ru.

0010-5082/07/4302-0212 © Springer Science + Business Media, Inc.

ume of 4.75 liters, in which 50–100 g charges in a lead shell were exploded. After ignition, temperature was measured by three thermometers placed in special pock-ets arranged on a circumference on the bomb surface. According to [1], the measurement error was 1–2%. The Bichel bomb had an even greater volume — 26 liters. It allowed one to test charges of considerable diameter and mass exceeding 100 g. This brought the ignition con-ditions closer to the practical situation especially when the charge was confined in a lead shell, which slowed down the expansion of explosion products. In British re-search laboratories, investigators used a liquid calorime-ter with a bomb volume of 124 cm^3, which made it possible to test charges of mass up to 15 g. Starting in 1930, similar studies were performed in Germany by Hide and Schmidt, well-known scientists known in the 1920s–1930s.

It is incorrect to compare data on the heat of ex-plosion obtained experimentally using these techniques. The heat of explosion is a nonconstant parameter and depends strongly on experimental conditions (the pres-ence of a shell, the expansion ratio of explosion prod-ucts, critical diameter). Of course, data obtained us-ing the same technique provide a correct information of the energy release of a particular commercial ex-plosive. To estimate the heat of explosion and the gas volume of commercial HEs, it is reasonable to use approximate rules to write the Mallard–Le Chatelier and Brinkley–Wilson equations of explosive decompo-sition (with an accuracy of 10–20%). The explosive-decomposition equation is used to calculate the heat of explosion and the volume of gaseous products and to evaluate the working capacity (blast effect). In this case, the error is of no significance since the working capacity is tested relative to the working capacity of a reference. For commercial HEs, detonability, the avail-ability of raw materials, cost, and operation automation are of fundamental importance. Of all issues involved in choosing commercial HEs, only the issue of detonability (decreasing the critical diameter) is of research inter-est. For emulsion HEs, it is solved by the addition of gas-generating agents (such as $NaNO_2$) or hollow glass spheres, which decompose to produce gas bubbles upon and increase the detonability of the HEs.

EXPERIMENTAL METHODS FOR DETERMINING THE HEAT OF EXPLOSION OF BRISANT HE

During the Second World War, more powerful HEs than widely used TNT and its mixture with ammonium nitrate were developed. The work of unprecedented scope carried out to develop special equipment during the Second World War and the subsequent period re-quired HEs of the maximum power at that time. RDX and its explosive compositions with TNT of various proportions (20/80 TNT/RDX, 36/64 TNT/RDX, and 50/50 TNT/RDX), depending on the required power, were used for this purpose. The impressive progress made in the development of new ordnance in the short period has motivated considerable interest in the devel-opment of powerful HEs up to date. Explosives are un-rivaled in ease of operation, reliability, and controlling ultrahigh pulsed-power sources. Extensive studies in the area of HEs have led to the development of methods for investigating fast processes, physicochemical and ex-plosive properties, chemical and physical stability, and detonability and detonation parameters. In the USSR and the USA (Livermore), facilities have also been de-signed to study the heat of explosion of brisant HEs for the purpose of obtaining information on the energetics and power of HEs. Great hopes were pinned on the informability of the technique of determining the heat of explosion because detonation involves energy release due to the decomposition of HE, and according to clas-sical detonation theory, $C = Q/V$ and $P = I/Q$ where C and p are the detonation velocity and pressure and Q is the energy released in the Chapman–Jouguet plane, which determines detonation parameters and brisance. HE by itself is an ultrahigh energy source.

Liquid calorimeters with 2- and 5-liter bombs were designed and produced at the Institute of Chemical Physics of the Russian Academy of Sciences (ICP RAS) [2]. The 5-liter bomb calorimeter made from a gun shaft was used most widely in laboratory exper-iments. The bomb calorimeter designed to study HE charges of mass up to 100 g is shown in Fig. 1 and a de-tailed description of the experiments is given in [3]. In the process of operation, the facility with a bomb capac-ity of 5 liters was subjected to methodical changes due to the automation of the experiments with computeriza-tion of the full minutes of the experiment [4]. Stirring with a mixer was eliminated, which reduced the heat equivalent of the facility.

At the Livermore National Laboratory (USA), a liquid calorimeter with a spherical bomb of volume 5.24 liters was designed and used to study the heat of explosion of brisant HEs in approximately the same years as at the ICP RAS [5]. A diagram of the liquid calorimeter is presented in Fig. 2. In studies of fluorine-containing HEs (whose molecule contains —$C(NO_2)F$ and —NF_2 groups), in the bomb of the Livermore Na-tional laboratory, ignition was performed in a gold shell, and in the bomb of the ICP RAS, in a brass shell, which contained a HE and a small amount of KOH. A correc-

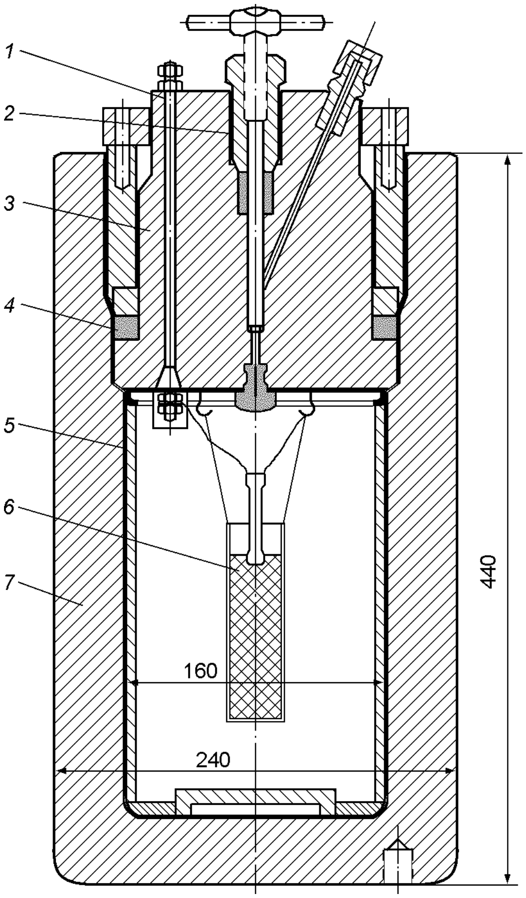

Fig. 1. Bomb calorimeter designed to study the heat of explosion of HE charges of mass up to 100 g (de-velopment of the ICP RAS): 1) electric lead; 2) valve;
3) self-sealed cover of the bomb; 4) Teflon or rubber seal; 5) changeable metal liners protecting the bomb walls from being damaged by debris; 6) HE charge;
7) bomb casing.

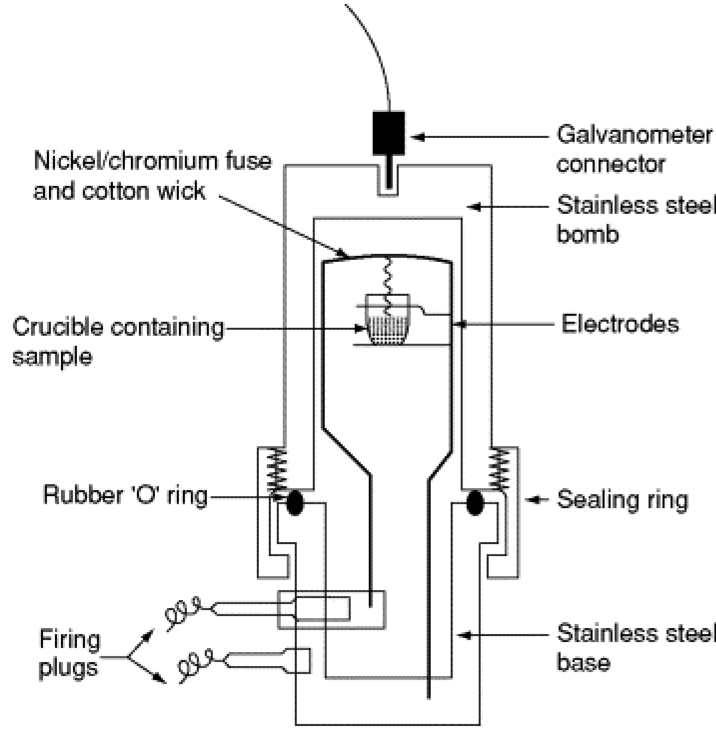

Fig. 2. Bomb calorimeter (development of the Liver-more National Laboratory): 1) thermistor sensor; 2) re-sistance thermometer; 3) mercury–glass thermometer;
4) calorimeter sleeve with edge; 5) polystyrene foam compression struts; 6) adjustment rope; 7) polystyrene foam insulator; 8) electrode; 9) knife type heater;
10) mixer; 11) bomb; 12) constant-temperature jacket.

tion was made for the interaction of aggressive detona-tion products with KOH. This correction was obtained and justified in studies of the CH_3NO_2 + XeF_2 system performed in a steel bomb and in a bomb whose in-ner surface was boarded coated with platinum. The calorimetric facilities of the ICP RAS and the Liver-more National Laboratory are close in test conditions, are comparable in measurement accuracy, and give close experimental data on heats of explosion. The results of calorimetric measurements of the heat of explosion of PETN and gas analysis of its detonation products obtained at the facility of the Livermore National Lab-oratory are given in [5]. For purity of the experiment (this is a very important methodical point), the det-onator also contained a PETN pellet of high density and 150 mg of powdered PETN around a wire bridge.

At the ICP, the same problem was solved by carrying out a blank run with ignition of a plurality (not less than 30) of blasting caps and a correction was made for the use of one blasting cap. In some experiments (for HEs of practical interest), initiation was performed by lead azide, which in the wet state (incapable of detona-tion) was placed at the end of a pressed HE charge and, then, during air exhaust from the bomb and its evac-uation, it lost moisture and initiated the charge. The experimental data for PETN were interpreted in [5] us-ing numerical hydrodynamic and thermodynamic cal-culations, which provided a quantitative solution of the problem of the energy distribution under the experi-mental conditions. During detonation of unconfined charges, the released energy remains predominantly in the products, which are compressed by the shock wave reflected from the bomb walls, and their state after compression differs significantly from the state at the Chapman–Jouguet point. During detonation of charges in massive shells, the energy is converted to the kinetic and internal energy of the shell and the products are ex-panded along the Chapman–Jouguet isentrope without being compressed by the shock wave. Since the deto-nation products formed from charges in massive shells are not loaded by a shock wave, the final composition of the products is close to the composition for the isen-tropic expansion of detonation products to the freez-ing temperature of 1500–1800 K. Subsequently, these findings were confirmed in studies [6, 7] of represen-tatives of various classes of HEs — ordinary CHON ni-tro compounds (nitromethane, TNT, and HMX of three densities), CHONF fluoro nitro compounds (bis-fluoro-dinitro-ethylformal), CHNF difluoro amino compounds [1,2-(difluoro-amino)propane], CNO-nonhydrogen sub-stances (benzotrifuroxane) and HNO nonhydrogen sub-stances (hydrazine nitrate/hydrazine system). The lat-ter system (known as Astrolite) with a variable hy-drazine content can have various densities and, hence, detonation velocities.

Basic research [5–7] has shown that the heat of explosion of brisant HEs is a variable parameter that depends on what gas expansion ratio the experimental measurements were made. Experimental values of the heat of explosion of commercial and brisant HEs reflect different processes. The heat of explosion of brisant HEs transferred through the bomb walls to the calorimetric liquid is the experimentally detected heat content of the gaseous detonation products expanding to a certain de-gree. Approximate methods for calculating the heat of explosion of brisant HEs using the standard thermo-chemical program with the ideal gas equation of state are described in [8]. The accuracy of these approximate calculation methods is far lower than the accuracy of the empirical methods of [9], whose error is 1.8%.

An energetic parameter of a HE is the maxi-mum heat of explosion, which reflects the HE structure through the enthalpy of formation and its chemical ele-ment composition. The maximum heat should be used to evaluate the limiting capabilities of a particular HE during transformation of chemical energy to thermal en-ergy. The maximum heat of explosion corresponds to the maximum possible heat effect of detonation of the given HE.

Kamlet and Jacobs [10] proposed a method for cal-culating the detonation velocity and pressure of brisant HEs using the maximum heat of explosion. They statis-tically treated calculated thermodynamic data for det-onation of various HEs. An analysis of the calculated compositions allowed them to formulate a generalized rule for estimating the detonation-product composition at the Chapman–Jouguet point: the main products are water, nitrogen, carbon dioxide, and condensed carbon. This rule is used to estimate the composition and deter-mine the number of moles of the gas-phase detonation products per 1 g of HE (N), the mean molar weight of detonation products (M), and the maximum possible heat of explosion (Q_{max}). The parameters Q_{max}, N , and M — constants for a given HE — are integrated into a generalized parameter, which, along with the HE density, is a regression parameter for the detonation ve-locity and pressure. The choice of the parameters Q_{max}, N , and M is quite reasonable. The method of Kamlet and Jacobs is the most accurate and simple and provides a high rate of obtaining results. Comparative indicators of its accuracy are given in [11].

The application of the Brinkley–Wilson rule to the estimation of the detonation velocity of brisant HEs leads to high inaccuracies. The use of this rule along with the empirical relation obtained by statistical pro-cessing of a small number of data linking pressure, spe-cific volume, and temperature results in large errors in the Voskoboinikov method [12]. Estimated detonation velocities of HEs of anomalous element compositions (nonhydrogen, fluorine-containing, and noncarbon) are almost 1.5 times higher than experimental values.

Detonation is a complex supersonic process of prop-agation of a shock wave followed by the chemical reac-tion. Experiments on heats of explosion only detects the heat content and composition of the gases in the stage of their isentropic expansion. Because of the high-density energy release, the detonation products gain a reserve of internal and kinetic energy that is spent to perform work. From the viewpoint of performing work, gaseous detonation products transfer a considerable portion of the potential chemical energy irrespective of the method of transfer — due to the shock wave or due to pres-sure variation during isentropic expansion. Detonation is characterized by the most intense shock-wave mecha-nism of energy transfer. The amount of energy released in the reaction zone rather than the heat of explosion is a characteristic of the detonation process but it cannot be determined experimentally.

CHEMICAL HEAT AND CALCULATION OF THE HEAT OF EXPLOSION

The development of physically justified equations of state for fluids (gaseous detonation products in the supercritical state in the chemical-reaction zone, where the differences between gas and liquid disappear) and for condensed nanocarbon phases (nanographite, nan-odiamond, and liquid carbon) [13] and the development of the corresponding thermodynamic calculation meth-ods also provide information on the element composi-tion of detonation products at the Chapman–Jouguet point and in the isentrope. In turn, knowledge of the element composition of the products allows the heat of explosive decomposition to be calculated as the differ-ence of the chemical constituent of the enthalpy for a definite expansion ratio and the enthalpy of formation of the starting HE. As is known, the detonation process and the work of detonation products in expansion are performed due to the part of the internal energy accumulated in the chemical bonds of HE molecules. This energy, released during chemical interaction, is converted to kinetic energy and the thermal and elas-tic constituents of the internal energy of the detonation products. Therefore, one can introduce the notion of chemical heat, which is defined as the difference of the chemical constituents of the enthalpies of the starting substance and the current state of the detonation prod-ucts.

In addition to intratomic energy and the energy of zero vibrations of chemical bonds in the molecule, the chemical constituent of the enthalpy of a condensed substance includes the energy of zero vibrations of in-termolecular bonds, which is usually called the elastic energy of the crystal lattice. Thus, the chemical compo-nent of the enthalpy of a condensed substance is equal to the enthalpy of this substance at zero ambient pressure and a temperature of 0 K. However, data on the en-thalpy at zero temperature are not available for all sub-stances, and the more so for HEs. Therefore, the specific chemical heat is calculated as the difference of the spe-cific enthalpies of the HE and the detonations products in the standard state at a pressure of 101,325 Pa and a temperature of 298.15 K, which is equal to the differ-ence of the standard specific enthalpies of formation of the HE and the products:

$$q_x = \sum_i n_i \Delta H^0_{f\,i} - \sum_j n_j \Delta H^0_{f\,j},$$

TABLE 1

Comparison of Calculated Heats of Explosion with Experimental Data

HE	Formula	ρ_0, g/cm^3	q_{exp}, cal/g [9]	q_{chem}, cal/g	dq, %
Nitro-guanidine	$CH_4N_4O_2$	1.58	820	798	2.9
TNT	$C_7H_5N_3O_6$	1.60	1030	990	3.8
HMX	$C_4H_8N_8O_8$	1.80	1300	1360	4.6
RDX	$C_3H_6N_6O_6$	1.70	1290	1260	2.4
Bensotri-furoxane	$C_6N_6O_6$	1.76	1290	1330	2.3
Tetryl	$C_7H_5N_5O_8$	1.69	1160	1200	3.4
Picric acid	$C_6H_3N_3O_7$	1.70	1010	1040	3.0
Diaminotri-nitrobenzene	$C_6H_5N_6O_6$	1.50	930	960	3.6

expansion ratio $V/V_0 \approx 2$ (in experiments, the charge is placed in a cylindrical metal shell). The blast heat Q_{blast} is the chemical heat of detonation products whose expansion is restricted by the walls of a bomb calorime-ter (in experiments, the charge is freely suspended to the bomb cover). Depending on the volume of the bomb, V_{bomb}/V_{ch} = 160–200, where V_{bomb} is the volume of the bomb and V_{ch} is the volume of the charge. The real chemical blast heat corresponds to the conditions under which the expanding detonation prod ucts reach

where the subscripts i and j refer to the reactants and products, respectively, n is the number of moles of the individual substance in 1 kg, and ΔH^0_f is the enthalpy of formation. As follows from the formula, the chemical heat depends only on the composition of the reactants and products and, hence, it can serve as a general char-acteristic of the product composition. Since the internal energy of the detonation products changes continuously during isentropic expansion, there is a certain expansion ratio for which the calculated value of the chemical heat is equal to the heat of explosion. The best agreement be-tween calculated chemical heat and experimental heats of explosive decomposition of brisant HEs is observed for an expansion ratio $V/V_0 = 2$, where V is the cur-rent volume of the detonation products and V_0 is their initial volume. Table 1 compares calculated chemical heats and experimental heats of explosion.

In [14], the heat of explosion is subdivided into the detonation heat (Q_{det}) and the blast heat (Q_{blast}). These quantities both represent the chemical heat (heat content) of detonation products for different expansion ratios. The detonation heat Q_{det} corresponds to an

the ambient temperature and pressure. The experimen-tal heats of explosion Q_{det} and Q_{blast} are only a small portion of the real blast heat of the explosion.

Methodical studies and the empirical method for calculating the heat of explosion [15–18] only give the appearance of the value and significance of the heat of explosion of powerful HEs since it is not used in calcu-lations. In practice, the blast effect is usually enhanced by adding high-calorific metal components to HEs (as a rule, the highest-calorific metal is aluminum). The addition of metal to brisant HEs results in a pressure decrease at the detonation front and its flatter decay, which increases the time of the pressure effect and en-hances the blast effect. In addition to the time of the pressure effect, the blast effect also depends on the vol-ume and composition of the gaseous detonation prod-ucts — the working medium that performs the blast effect. Qualitatively, the volume, the gas composition, and calorific value of the metallized additives are es-timated based on the equation of explosive decomposi-tion with the maximum energy release — a constant for metallized compositions. In practice, the effect of metallized compositions is characterized by generally ac-cepted parameters other than the energy release. Met-allized compositions are commonly used in brisant-blast warheads (fragmentation warheads, underwater explo-sions). During operation, the brisant effect is enhanced by the blast (demolition) effect due to pressure varia-tion during the chemical interaction of the metal with the detonation products under isentropic expansion.

Experimental data on the heats of explosion of brisant HEs provide little information for an analysis of high-speed chemical processes.

WORKING CAPACITY OF COMMERCIAL AND BRISANT HEs

The action of an explosion is subdivided into blast and brisant effects. The blast effect is determined as the total working capacity. The definition of the working capacity of HEs was given by Belyaev [19]. The total work of an explosion is usually calculated by the formula

$$A = Q\left[1 - \left(\frac{V_{in}}{V_{fin}}\right)^{n-1}\right],$$

where V_{in} and V_{fin} are the initial and final volumes of the explosion products, n is the polytropic exponent, A is the total work, and Q is the heat of the explosion. The total work of explosion is calculated using table and calculated values of Q and V_{in} and the constant value n = 1.25. This quantity is called the ideal work of explosion and is denoted by A_{id}. It is included in the reference literature [20] and characterize the work-ing capacity of HEs (A_{id} is necessarily included in the technical specifications of HEs). The working capac-ity of a HE is characterized by the heat of explosion. As a result, we have an energetic characterization of the working capacity. In recent years, the notion of the relative working capacity f was introduced. The quantity f is defined as the ratio of the work of ex-plosion of 1 kg of a new HE to the work of explo-sion of 1 kg of the reference HE — 79/21 AN/TNT. In the development of a new HE, the heat of explosion, pressure, and the parameter f are usually calculated beforehand and are then tested experimentally. Since thermodynamic methods are inappropriate for calculat-ing the detonation parameters and working capacity of commercial HEs in view of their nonideal detonation processes, such calculations use simple empirical ex-press methods and approximate rules for writing the Brinkley–Wilson equations of HE decomposition. From the HE decomposition reactions, one obtains the values of Q and $V_0 = 22.4N$, where N is the number of moles of gaseous products. The application of the approxi-mate rules of HE decomposition to the calculation of the working capacity of commercial HEs is quite jus-tified and reasonable since new HEs are tested experi-mentally and compared with the data for the reference HE — 79/21 AN/TNT. It should be noted that the er-ror of such experiments (cavity diameter, charge depth, cavity volume, and ground density) is comparable with the calculation error. Afanasenkov [21] gives the follow-ing formula to estimate the relative working capacity of commercial HEs:

$$f = \left(\frac{Q}{1031}\right)^{0.75}\left(\frac{V_0}{893}\right)^{0.25}$$

where values 1031K and 893K are the table data for 79/21 AN/TNT ⅔ cylinder (reference HE); Q in kcal/kg; V_0 in liters/kg. The brisant effect or, by the nomenclature of Belyaev and Sadovskii, the local (short-range) action of explosion was evaluated by them from the compres-sion of copper crushers on a device for brisance mea-surements. In recent years, use is made of the con-cept of the propellant performance of HEs, which is directly related to the blast effect. The propellant per-formance is determined experimentally using two gen-erally accepted techniques. One is the end acceleration of plates, in which a plate acceleration curve and the velocity of approach to a place at a strictly fixed dis-tance are recorded. In this method, the measurement base is limited because of the plate curvature during acceleration. The second technique is the cylinder test or the domestic T-20 technique: the velocity of a cylin-drical shell of annealed copper is recorded at a distance of 5 mm (R/R_1 = 1.33) and 19 mm (R/R_1 = 2.24) is fixed. The power of a HE is judged by the relative pro-pellant capacity. For the end acceleration, the reference is HMX and the propellant performance of its charge at a density of 1.875 g/cm^3. In the cylinder test, the reference is the kinetic energy of a shell accelerated by the detonation products of a 36/64 TNT/RDX charge. The propellant performance is a complex characteristic of the power of a HE that is determined by its structure through the enthalpy of formation, the chemical element composition, and density. The last characteristic is very important since all detonation parameters are propor-tional to it. The density of a HE single crystal depends on the element composition, structure, the type of crys-tal lattice, and the packing coefficient. The density of the HE charge determines the energy release density in unit volume of the reaction zone and the high energy-release power at the front. Usually, the power of a HE is judged by the detonation velocity. Studies of the pro-pellant performance have shown that the detonation ve-locity cannot be a measure of the HE power. It suffices to note that HEs such as methylenedinitramine (MED-INA) and ethylenedinitramine (EDNA) have a low level of propellant performance at a high detonation velocity

(≈9000 m/sec). Explosives such as hydrazine nitroform and dinitroguanidine, which are similar and even supe-rior to HMX in detonation velocity ($D \approx 9200$ m/sec), are inferior to it in propellant performance. Measure-
ments using the end acceleration method have shown that dinitroguanidine at a charge density of 1.81 g/cm^3 (single crystal density of 1.88 g/cm^3) has a relative pro-pellant capacity of 95.3%, which is close to the propel-lant capacity of RDX of 95.7%. The issue of the propel-lant performance is of extreme theoretical and especially practical importance. This problem, together with the problems of the power, explosion safety, and quality of the working medium, requires a separate consideration.

CONCLUSIONS

1. The techniques used to determine the heat of explosion of commercial and brisant HEs was analyzed.
2. It was shown that the heat of explosion is not a constant of a HE and depends on the expansion ratio of the detonation products.
3. The heat of explosion is calculated by solving the thermodynamic problem for various expansion ratios.
4. A comparison was performed of experimental and calculated heats of explosion of brisant HEs.
5. The heat of explosion is reasonably included in the energetic characterization of the working capacity of commercial HEs, but it has no relation to the power of brisant HEs, which is determined by the propellant capacity. Even the detonation velocity is not a measure of the power of HEs.

This work was supported by the Russian Founda-tion for Basic Research (Grant No. 04-03-32873).

REFERENCES

1. A. Ya. Apin and A. F. Belyaev, “Experimental determi-nation of the heat of explosion,” in: *Physics of Explosion* (collected scientific papers) [in Russian], No. 2 (1953), pp. 3–26.
2. Yu. A. Lebedev, G. G. Lipanin, V. I. Pepekin, and A. Ya. Apin, “Thermochemical study of individual ex-plosives and their mixtures,” *Vzryvnoe Delo*, No. 52/9, 80–90 (1963).
3. A. Ya. Apin, N. F. Velina, and Yu. A. Lebedev, “Com-plete utilization of the energy of explosion,” *Prikl. Mekh. Tekh. Fiz.*, No. 5, 96–106 (1962).
4. V. I. Pepekin and A. S. Smirnov, “Heats of explosive de-composition of mixtures containing aluminum and ener-getic substances,” *Khim. Fiz.*, **20**, No. 11, 78–80 (2001).
5. D. L. Ornellas, J. H. Carpenter, and S. R. Gunn, “Det-onation calorimeter and results obtained with pentaery-thritol tetranitrate (PETN),” *Rev. Sci. Instr.*, **37**, 907 (1966).
6. D. L. Ornellas, “The heat and products of det-onation of cyclotetramethylene-tetranitramine, 2,4,6-trinitrotoluene, nitromethane, and bis[2,2,2-dinitro-2-fluoroethyl]formal,” *J. Phys. Chem.*, **72**, 2390 (1968).
7. D. L. Ornellas, “The heat and products of detonation in a calorimeter of CNO, HNO, CHNF, CHNO, CHNOF, and CHNOSi explosives,” *Combust. Flame*, **23**, 37–46 (1974).
8. R. A. Fifer and J. B. Morris, “Approximate techniques for the calculation of heats of explosion using ther-mochemical computer codes,” *Thermochim. Acta*, **237**, No. 2, 375–389 (1994).
9. I. V. Pepekin, M. N. Makhov, and Yu. A. Lebedev, “Heats of explosive decomposition of individual HEs,” *Dokl. Akad. Nauk SSSR*, **232**, No. 4, 852–855 (1972).
10. M. J. Kamlet and S. J. Jacobs, “Chemistry of detona-tion. Simple methods for calculation detonation prop-erties of CHNO-explosives,” *J. Chem. Phys.*, **48**, 23 (1968).
11. I. V. Pepekin and S. A. Gubin, “Methods of calculating the detonation parameters of explosives,” *Khim. Fiz.*, **22**, No. 9, 72–97 (2003).
12. A. Ya. Apin and I. M. Voskoboinikov “Calculation of the detonation wave parameters of condensed explosives,” *Prikl. Mekh. Tekh. Fiz.*, No. 4. 54–55 (1960).
13. S. B. Viktorov, S. A. Gubin, I. V. Maklashova, and V. I. Pepekin, “Prediction of detonation characteristics of nonhydrogen HEs,” *Khim. Fiz.*, **24**, No. 12, 22–45 (2005).
14. L. P. Orlenko, *Physics of Explosion* [in Russian], Fiz-matlit, Moscow (2004).
15. M. N. Makhov, “Determination of the energy content of individual HEs,” *Khim. Fiz.*, **19**, No. 6, 52–56 (2000).
16. M. N. Makhov, “Heat of explosive decomposition of aluminized explosives,” *Khim. Fiz.*, **19**, No. 9, 83–87 (2000).
17. M. N. Makhov, “A simplified method for predicting the heat of explosion of aluminized high explosives,” in: *Proc. 35th Int. Annu. Conf. ICT*, Karlsruhe, FRG, (2004), pp. 3/1–3/8.
18. M. N. Makhov, “Estimation of the heat of explosive de-composition of aluminized HEs,” in: *Progress in Special Chemistry and Chemical Technology* (collected scientific papers) [in Russian], Izd. Mendeleev Ross. Gos. Tech. Univ. (2005), Part 2, pp. 173–177.
19. A. F. Belyaev, “On the total work of explosion,” in: *Physics of Explosion* (collected scientific papers) [in Russian], No. 2 (1953), pp. 27–42.
20. *List of Recommended Commercial Explosives and Explo-sion and Control Devices* [in Russian], Nedra, Moscow (1987).
21. A. N. Afanasenkov, L. I. Kotova, and B. N. Kukib, “Strength of commercial explosives,” *Combust., Expl., Shock Waves*, **37**, No. 3, 349–358 (2001).
22. International, Diamond, “136-Page Fleetrite Parts Catalog”, **87**, No. 3, 87-107.